

On-Chip Sleep Quality Prediction using Digital Biomarkers & *Neural Networks*

Predicting next-night sleep disruption from wearable heart-rate signals — and shrinking the model small enough to run directly on a wristband chip, with no cloud.

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THE PROBLEM

Poor sleep is linked to cardiovascular disease, stress and cognitive decline. The clinical gold standard — polysomnography — needs a sleep lab and cannot run continuously. **Wearables already capture the signals we need**, but the intelligence usually lives in the cloud.

We predict **WASO** (Wake After Sleep Onset) — a key marker of fragmented sleep — one night ahead, and run that prediction **entirely on the device chip**.

OBJECTIVES

- Deploy a sleep-prediction model on **low-power embedded hardware**.
- Design a lightweight architecture for **TensorFlow Lite Micro**.
- Build a pipeline: training → conversion → **INT8 quantization**.
- Implement embedded inference firmware for **Arduino-class chips**.
- Validate the quantized model against the floating-point baseline.

DATASET & FEATURES

Seven signals per day, over a 7-day wearable window, feed the model:

- LF/HF ratio & RMSSD** — heart-rate variability
- Steps, accelerometer, gyroscope** — physical activity
- ISI & WHOQOL** — insomnia & quality of life

Clinical questionnaires collected in the study (ISI & WHOQOL feed the model; the rest characterise the cohort):

ISI · insomnia

WHOQOL · quality of life

PHQ-9 · depression

GAD-7 · anxiety

KNHANES · stress

A **subject-wise split** keeps each participant in only one of train/test — removing leakage present in the original study.

SIGNATURE RESULT – MODEL COMPRESSION

50MB → **7.5KB**
Original LSTM (Cloud / GPU only) → Quantized 1D-CNN (Runs on the chip)

A **~6,600× smaller** model — fits a 256 KB microcontroller.

METHOD – FROM LSTM TO CHIP

- Redesign:** recurrent LSTM → compact **1D-CNN** (recurrent ops unsupported on bare-metal MCUs).
- Train:** 1,425 parameters, binary cross-entropy, EarlyStopping on validation AUROC.
- Convert:** Keras → TensorFlow Lite with a representative calibration set.
- Quantize:** full **INT8** — weights & activations to 8-bit integers.
- Deploy:** C header + firmware on Arduino Nano 33 BLE / ESP32 / STM32.

QUANTIZATION IMPACT

Model	AUROC	Accuracy
Floating-Point CNN	0.813	73.1%
INT8 Quantized CNN	0.803	72.8%

Full integer quantization cost only **0.31% accuracy** — the compact model keeps almost all of its predictive power while becoming small enough to flash onto a wearable chip.

On-device: ~8 ms per inference, ~25 KB RAM of 256 KB used.

0.87

Best AUROC (ensemble)

7.5KB

On-chip model size

~8ms

Inference latency

8

Models benchmarked

BENCHMARK – TOP MODELS

Model	AUROC	Acc.	Prec.
Ensemble ★	0.870	0.809	0.858
XGBoost	0.866	0.782	0.881
LSTM	0.856	0.718	0.923
Transformer	0.854	0.747	0.877
TCN	0.854	0.771	0.840
GRU	0.850	0.709	0.951
Random Forest	0.845	0.762	0.865
ARIMA (base)	0.485	0.613	0.705

Subject-wise split. The **LSTM + XGBoost ensemble** wins; all neural nets beat ARIMA decisively.

EXPLAINABILITY

SHAP (global) and **LIME** (local) independently agree on the top predictive drivers: **LF/HF ratio, insomnia severity, quality of life** and **RMSSD** — confirming the model relies on genuine physiology, not spurious correlations.

KEY CONTRIBUTIONS & NEXT STEPS

- Fixed a **data-leakage** flaw via subject-wise splits.
- Added **SHAP + LIME** dual explainability.
- Introduced an **LSTM + XGBoost ensemble** — best AUROC.
- Delivered a full **train-to-chip TinyML pipeline**.

Next: hardware trials, a live PPG sensor for real-time HRV, and clinical-data validation.